

AI Token Usage & Cost Baseline

Sections 4 & 5 · Observed Cost vs Unit Economics · Analysis & Improvement Plan

Date: 2026-05-10

Model: claude-sonnet-4-6 (Sonnet 4.6)

Currency: USD

1 LU = \$0.25

⚠ Partial — 4 actions pending observed data

1

Executive Summary

Model: claude-sonnet-4-6

This report captures the **first observed AI cost baseline** for SnapNotes. Nine of thirteen AI actions have been measured using production-equivalent token counts and Sonnet 4.6 pricing. Four actions (Homework Check, Diagram, Walkthrough, Flashcard Batch) have no observed data and remain on reference-cost estimates.

Headline finding: Every measured action came in significantly below the unit economics reference cost — ranging from –59.5% to –93.4% cheaper than modelled. This means the original pricing model was *consistently conservative*, and the actual gross margins across all measured actions are healthy.

Caution: These are single-sample observations, not averaged production logs. Variance across real user sessions may differ from these baseline measurements. Before locking this as the optimisation reference, production-period data should be collected and averaged.

9 / 13

Actions with observed
data

–59% to
–93%

Delta vs reference cost
(all measured)

84% – 97%

Gross margin range
(measured actions with
LU data)

4

Actions with no LU
pricing entry

Sonnet 4.6

Production model
(default)

Model note: All token costs in this report use Sonnet 4.6 rates: Input \$3.00/M, Output \$15.00/M, Cache Read \$0.30/M, Cache Write \$3.75/M. If the production model changes to Haiku 4.5 or Opus 4.7, all cost figures and margins will need to be recalculated.

2 Current Token Usage Baseline (Source: Doc §4)

⚠ 4 of 13 actions have no observed data. Rows highlighted in amber are TBD — only the unit economics reference cost exists for these actions. Do not use reference costs as a proxy for optimisation planning.

Full observed data Partial / TBD data — needs measurement

AI Action	Total Tokens	Input Tokens	Output Tokens	Cache Read	Cache Write	Image Fee	Observed Cost	Notes
Course Gen (Text)	17,142	2,837	14,305	0	0	\$0.00	\$0.2231	Text content input; highest output token count observed
Course Gen (Upload)	10,944	5,694	5,250	0	0	\$0.00	\$0.0958	20–30 min uploaded file; structured input → shorter output
Homework Check (Smart)	2,261	575	1,686	0	0	\$0.00	\$0.0270	Decompose + solve pipeline; far fewer tokens than modelled
Prepare Guide	8,181	1,395	6,786	0	0	\$0.00	\$0.1060	Single call; output lower than 16K reference assumption
Homework Check	TBD	TBD	TBD	TBD	TBD	TBD	TBD	No observed data — reference: \$0.237
Practice Session	10,698	6,698	4,000	0	0	\$0.00	\$0.0801	Generate + tutor turns; fewer turns than 3-turn reference
Diagram (with image)	TBD	TBD	TBD	TBD	TBD	incl. Recraft	TBD	No observed data — reference: \$0.175 (incl. \$0.04 Recraft)
Exam Generation	6,426	3,888	2,538	0	0	\$0.00	\$0.0497	2.5 question batches; strong efficiency
Lesson Expansion	839	310	529	0	0	\$0.00	\$0.0089	Single call; lowest token count of all measured actions
Chat (Prepare Guide)	317	14	303	0	3,024	\$0.00	\$0.0159	Cache write active — \$0.30/M rate applied to write tokens
Walkthrough	TBD	TBD	TBD	TBD	TBD	TBD	TBD	No observed data — reference: \$0.070
Chat (Course)	1,677	1,387	290	0	0	\$0.00	\$0.0085	Very short output; cheapest measured action

AI Action	Total Tokens	Input Tokens	Output Tokens	Cache Read	Cache Write	Image Fee	Observed Cost	Notes
Flashcard Batch	TBD	TBD	TBD	TBD	TBD	TBD	TBD	No observed data — reference: \$0.027

Observed Cost = (Billable Input / 1,000,000 × \$3.00) + (Output / 1,000,000 × \$15.00) + (Cache Read / 1,000,000 × \$0.30) + (Cache Write / 1,000,000 × \$3.75) Billable Input = Input Tokens – Cache Read Tokens

3 Comparison Against Unit Economics Model (Source: Doc §5)

Overestimated — reference cost was higher than actual (good: cheaper than planned)
 Needs investigation — no observed data yet

AI Action	Ref Cost (Sonnet 4.6)	Observed Cost	Delta \$	Delta %	LU / Action	Revenue @ LU	Gross Margin	Status
Course Gen (Text)	\$0.633	\$0.2231	– \$0.4099	–64.8%	10.0 LU	\$2.50	91.1%	Overestimated
Course Gen (Upload)	\$0.633	\$0.0958	– \$0.5372	–84.9%	10.0 LU	\$2.50	96.2%	Overestimated
Homework Check (Smart)	\$0.349	\$0.0270	– \$0.3220	–92.3%	4.0 LU	\$1.00	97.3%	Overestimated
Prepare Guide	\$0.264	\$0.1060	– \$0.1580	–59.9%	3.0 LU	\$0.75	85.9%	Overestimated
Homework Check	\$0.237	TBD	TBD	TBD	2.0 LU	\$0.50	TBD	Needs investigation
Practice Session	\$0.198	\$0.0801	– \$0.1179	–59.5%	2.0 LU	\$0.50	84.0%	Overestimated
Diagram (with image)	\$0.175	TBD	TBD	TBD	3.0 LU	\$0.75	TBD	Needs investigation
Exam Generation	\$0.137	\$0.0497	– \$0.0873	–63.7%	2.0 LU	\$0.50	90.1%	Overestimated
Lesson Expansion	\$0.135	\$0.0089	– \$0.1261	–93.4%	TBD	TBD	TBD	Overestimated
Chat (Prepare Guide)	\$0.082	\$0.0159	– \$0.0661	–80.6%	0.5 LU	\$0.125	87.3%	Overestimated
Walkthrough	\$0.070	TBD	TBD	TBD	TBD	TBD	TBD	Needs investigation
Chat (Course)	\$0.027	\$0.0085	– \$0.0185	–68.5%	0.5 LU	\$0.125	93.2%	Overestimated
Flashcard Batch	\$0.027	TBD	TBD	TBD	1.0 LU	\$0.25	TBD	Needs investigation

Delta \$ = Observed current cost – Unit economics reference cost
 Delta % = Delta \$ / Unit economics reference cost
 Gross Margin = (Revenue – Observed current cost) / Revenue
 Revenue = LU/Action × \$0.25

4 Analysis: What the Data Shows

1 Reference model was consistently conservative across the board

Every single measured action came in below its Sonnet 4.6 reference cost — none exceeded it. The original unit economics model was built on worst-case token assumptions (e.g. 16K output for Prepare Guide, 3 full tutor turns for Practice). Real usage produces shorter, more focused outputs. This is good news for margins but means the pricing model needs updating to reflect actual behaviour.

2 Gross margin reality is far better than the reference implied

The unit economics reference showed Course Generation at –153% gross margin (charging 1 LU for a \$0.63 action). With the actual LU pricing (10 LU = \$2.50 revenue) and the observed cost of \$0.22, the real margin is 91.1%. Similarly, HW Smart was modelled at –39.6% margin but is actually 97.3%. The LU pricing table is well-calibrated for revenue even if the cost reference was overstated.

3 Biggest outliers: Lesson Expansion (–93.4%) and HW Smart (–92.3%)

Both are single-call actions where the reference assumed significantly more output. Lesson Expansion at 839 total tokens (\$0.0089) is 15× cheaper than its \$0.135 reference. This suggests the reference was built from prototype-era token counts that never reflected final prompt design. These two actions have the largest optimisation "already done" — the actual implementation is already very lean.

4 Course Generation: text vs upload input diverge significantly

Text-content generation produced 14,305 output tokens (\$0.2231), while upload-file generation produced only 5,250 output tokens (\$0.0958) — a 57% cost difference for the same 10 LU charge. This is the single largest observed cost lever: if text-based inputs could be pre-structured to match the efficiency of file-based inputs, the average Course Generation cost would drop materially.

5 Chat (Prepare Guide) has cache write active — Chat (Course) does not

Chat (Prepare Guide) logs 3,024 cache write tokens, which at \$3.75/M adds a small premium over input tokens. However, if subsequent calls hit the cache, the \$0.30/M read rate would significantly reduce follow-up costs. Chat (Course) has zero cache activity — this is an untapped optimisation for multi-turn course chat sessions.

6 Lesson Expansion and Walkthrough have no LU pricing entry

Lesson Expansion at \$0.0089 per call generates real AI cost but has no corresponding LU charge defined in the pricing table. This means it is currently either free to the user or bundled invisibly into another LU charge. This is a revenue gap that should be resolved before billing goes live. Walkthrough has the same issue — no LU mapping and no observed data.

5 Fallbacks & Untested Actions

⚠️ **These 4 actions have no observed cost data.** The reference costs below come from the unit economics model only and may be as overestimated as the measured actions (which averaged –73% vs reference). Do not treat these as confirmed production costs.

Action	Reference Cost	LU / Action	Revenue @ LU	Pipeline Notes	Risk	Priority to Measure
Homework Check	\$0.237	2.0 LU	\$0.50	Solve + read + grade pipeline (multi-step). If actual mirrors HW Smart pattern, may be \$0.02–\$0.05.	Medium	High — high user frequency
Diagram (with image)	\$0.175	3.0 LU	\$0.75	Classify + schema + Recraft image call. Recraft flat fee (\$0.04) is the floor. Model tokens may be small.	High	High — external fee risk
Walkthrough	\$0.070	TBD	TBD	Single call, step-by-step. No LU pricing entry. Revenue unknown until LU is assigned.	Unknown	Medium — no revenue model yet
Flashcard Batch	\$0.027	1.0 LU	\$0.25	Single call, batch output. Reference is already low; likely accurate or below.	Low	Low — low-cost, low-risk action

Diagram risk note: The Diagram action is the only one with a *mandatory external API fee* (Recraft image generation at ~\$0.04 flat). Even if model token costs come in well below the \$0.135 token reference, the floor cost is \$0.04 regardless of token efficiency. This makes Diagram the one action where optimisation cannot fully eliminate costs — only reduce the token component.

6 What Can Be Improved

HIGH Collect missing data for 4 untested actions before locking any optimisation baseline

Homework Check, Diagram, Walkthrough, and Flashcard Batch have no observed data. Any optimisation strategy built without this data is incomplete. Instrument each action with token logging (input, output, cache read/write) and aggregate at least 20–30 real user calls per action before declaring the baseline final.

Impact: Closes the 31% gap in baseline coverage. Prerequisite for all other optimisation work.

HIGH Assign LU pricing for Lesson Expansion and Walkthrough

Both actions have confirmed or implied AI costs (\$0.0089 for Lesson Expansion) but no LU entry in the pricing table. They are either being given away for free or silently bundled. At \$0.0089/call and 84%+ margins on comparable actions, Lesson Expansion could be priced at 0.5 LU (\$0.125 revenue) for a gross margin of ~93%. Walkthrough needs its LU value set before any cost model is complete.

Impact: Revenue uplift on currently untracked AI spend. Fixes a gap in the unit economics model.

HIGH Prompt compression for Course Generation (Text) — output token reduction

Course Generation (Text) produced 14,305 output tokens — the highest of any measured action and the primary cost driver at \$0.2231. The upload-based variant produced only 5,250 output tokens for the same task at 57% lower cost. Applying structured output constraints, JSON schema enforcement, or output length caps to the text-based pipeline could reduce output tokens by 30–50%, bringing cost closer to the upload variant.

Estimated saving: –\$0.07 to –\$0.11 per text-based course generation (30–50% of \$0.2231).

MEDIUM Extend prompt caching to Course Chat and Prepare Guide generation

Chat (Prepare Guide) already writes 3,024 cache tokens, but Chat (Course) has zero cache activity. For multi-turn course chat sessions where the system prompt and course content are stable across turns, adding a cache write on the first turn and cache read on subsequent turns would reduce per-turn input cost from \$3.00/M to \$0.30/M for the cached portion. Similarly, the Prepare Guide generation call (\$0.1060) could benefit from caching the guide schema prompt.

Estimated saving on Chat (Course): –40–60% per follow-up turn in multi-turn sessions.

MEDIUM Content-type routing: pre-process text inputs to match upload pipeline efficiency

The 57% cost gap between text (\$0.2231) and upload (\$0.0958) course generation is the largest single cost lever in the measured data. If raw text inputs could be pre-processed (chunked, summarised, or structured) before being passed to the generation call, the effective output token count may approach the upload baseline. This would require a pre-processing step but could save ~\$0.13 per text-based course generated.

Estimated saving: –\$0.12 per text-based course (if output is reduced to upload-equivalent levels).

MEDIUM Replace single-sample observations with averaged production data

All current observed costs are single-sample measurements, not production averages. A single course generation run may not be representative — content length, complexity, and user behaviour vary. Before treating these figures as the optimisation baseline, collect at least 50 representative requests per action from production (or staging with realistic data) and use the p50 (median) as the baseline cost.

Risk reduction: Prevents optimising against a non-representative baseline. Required for credible post-optimisation comparison.

LOW Update the unit economics reference model to reflect actual observed costs

The current reference model overestimates costs by 60–93% across all measured actions. Once the baseline is finalised (after collecting missing data and averaging production calls), the unit economics reference table should be updated to use observed costs as the new planning baseline. This ensures future pricing decisions are built on reality rather than conservative assumptions.

Impact: More accurate LU pricing decisions and margin forecasting going forward.

7 Model & Pricing Reference

Sonnet 4.6 Token Rates (used for all calculations)	
Input (billable)	\$3.00 / 1M tokens
Output	\$15.00 / 1M tokens
Cache read	\$0.30 / 1M tokens
Cache write	\$3.75 / 1M tokens

LU Pricing (from official pricing table)	
1 Learning Unit (LU)	\$0.25
Create Course	10.0 LU → \$2.50
HW Smart	4.0 LU → \$1.00
Study Guide / Diagram	3.0 LU → \$0.75
HW Standard / Practice / Exam	2.0 LU → \$0.50
Flashcards	1.0 LU → \$0.25
Chat	0.5 LU → \$0.125

⚠ **Lesson Expansion and Walkthrough are not in the LU pricing table.** Both actions have real AI costs (\$0.0089 observed for Lesson Expansion; \$0.070 reference for Walkthrough) but no defined LU charge. These must be assigned LU values before the billing system goes live. Until then, gross margin and revenue cannot be calculated for these two actions.